

Problem 4

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Problem 1 Find all real numbers which satisfy each of the equations. In the previous problem, we used θ to denote the radian measure of the angle. However, we can use any variable to represent the angle measure. For example, in part a, x is the variable representing the radian measure of the angle.

(a) $\cos(x) = 1$

Explanation. This is asking for the collection of all angles such that cosine of that angle equals 1.

The unit circle shows that one such angle is 0 (since $\cos(0) = 1$). There is a slight trick here: since cosine has period 2π we actually have $\cos(0+2\pi n) = 1$ for every integer n . In summary, $x = 2\pi n$, where n is any integer, gives all the solutions to this equation.

(b) $\sin(3\theta) = \sqrt{3}/2$ for $0 \leq \theta \leq 2\pi$

Explanation. Finding all numbers θ with $0 \leq \theta \leq 2\pi$ that satisfy $\sin(3\theta) = \sqrt{3}/2$ is a bit tricky. We first perform a useful trick from algebra — variable substitution.

Let $x = 3\theta$. So, we are trying to find all numbers x such that $\sin(x) = \sqrt{3}/2$ for $0 \leq x/3 \leq 2\pi$. Then $x = \frac{\pi}{3} + 2\pi n$ or $x = \frac{2\pi}{3} + 2\pi n$ for n any integer as long as $0 \leq \theta \leq 2\pi$. Since $x = 3\theta$, we can solve for θ to obtain $\theta = \pi/9 + (2\pi n)/3$ or $\theta = (2\pi)/9 + (2\pi n)/3$, where n is again any integer as long as $0 \leq \theta \leq 2\pi$. We are only looking for solutions of θ in $[0, 2\pi]$, and so our solutions are

$$\theta = \frac{\pi}{9}, \frac{2\pi}{9}, \frac{7\pi}{9}, \frac{8\pi}{9}, \frac{13\pi}{9}, \frac{14\pi}{9}.$$